

Climate Organizes but Does Not Determine Wildfire Development

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One-Sentence Summary

Fire VASE reveals that climate shifts wildfire developmental opportunity, while prior fire state and unmeasured landscape context shape which opportunity is realized.

Abstract

Wildfire analyses often compare final burned area, duration, or average spread, even though fires with similar outcomes can develop through different sequences of growth, quiescence, reactivation, and termination. We introduce Fire VASE as a developmental representation that converts each daily fire history into a comparable geometric profile. Across 278,569 FIRED events from 2000-2021, 237,235 fires have complete daily centroid gridMET temperature, vapor pressure deficit (VPD), wind, precipitation, humidity, fuel moisture, fire-danger, evapotranspiration, and solar-radiation exposure. Fire VASE shows that wildfire histories vary along recurring but continuous gradients of timing, persistence, concentration, pulse structure, reactivation, and termination. Climate is associated with these gradients: high-VPD, hot, dry-fuel, and high fire-danger exposure groups differ in developmental-neighborhood prevalence and median profile shape. However, blocked validation is weak, region-season anomaly diagnostics do not rescue deterministic prediction, and similar centroid climate can yield divergent VASE forms. Leakage-safe next-day models show that current developmental state improves interpretation of growth response beyond climate alone, whereas climate-state interactions remain suggestive rather than definitive under conservative blocking. Fire VASE therefore changes the climate question: climate organizes developmental opportunity, but it does not uniquely determine realized wildfire development.

Introduction

Climate is a first-order constraint on wildfire activity, but burned area is not the same as development. A fire can reach a final area through an early burst, steady accumulation, late surge, repeated pulses, or reactivation after quiescence. These histories matter because the timing and concentration of burning shape exposure, ecological effects, and opportunities for response. Yet continental fire archives are often reduced to final size, duration, or aggregate spread rate.

Fire VASE was designed to preserve this missing developmental information. It maps developmental time to vertical position and cumulative burned area to ring width, producing a comparable object for every observed fire history. Climate is then projected onto the object rather than used to define the coordinate system. This distinction matters: a representation can reveal how climate is translated into growth without assuming that climate alone prescribes the resulting form.

Here we rebuild the Fire VASE analysis around one question: how does climate organize wildfire developmental opportunity? We use daily FIRED fire histories derived from MODIS burned area event delineation [1,2] and daily gridMET climate fields [3]. The population-wide table now includes daily centroid gridMET maximum temperature, minimum temperature, VPD, wind speed, precipitation, relative humidity, specific humidity, 100-hour and 1000-hour fuel moisture, energy release component, burning index, reference evapotranspiration, potential evapotranspiration, and solar radiation for 237,235 climate-complete fires. Perimeter, active-burned-area, and perimeter-extension attribution remain a separate exposure product and are treated according to their actual coverage rather than used as the main inferential basis.

Results

Fire VASE preserves developmental differences hidden by final outcomes

Simple final summaries can hide visibly different daily growth histories. Figure 1 shows real events with contrasting temporal allocation: some accumulate most area early, others grow steadily, others grow late, and others develop through multiple pulses. The corresponding VASEs preserve those differences in one visual grammar. This establishes Fire VASE as the instrument for the climate analysis rather than as the paper's endpoint.

Wildfire histories vary along recurring developmental gradients

Observed fires occupy recurring neighborhoods that are best interpreted as landmarks along continuous gradients. Figure 2 uses descriptive labels such as skinny persistent, compact steady, late surge, front-loaded plateau, and multi-pulse complex. These categories help communicate form, but they are not hard biological types. The important result is that developmental variation can be described in terms of timing, persistence, concentration, pulse structure, reactivation, and termination, while still preserving absolute-scale outcomes separately.

Climate shifts the probability of developmental forms

Daily centroid gridMET climate varies systematically across the Fire VASE representation. Event-mean VPD, maximum temperature, humidity, fuel moisture, precipitation, and fire-danger summaries are associated with interpretable developmental responses, including front-loaded growth, late growth, pulse count, reactivation, and the dominant VASE gradient. Composite VASEs across VPD terciles show that high- and low-VPD groups differ in where normalized growth is allocated through developmental time. Developmental-neighborhood prevalence also shifts across VPD groups.

The predictive limit is equally important. In conservative blocked validation, the best transferable event-level representation is **core event means**, with median held-out R^2 of 0.349 across developmental responses. Region-season anomaly diagnostics do not outperform core event means, comprehensive event means do not outperform core event means, and temporally resolved exposure summaries do not outperform core event means in the median blocked comparison. These results support a probabilistic statement: climate redistributes fires across developmental possibilities. They do not support the stronger statement that climate assigns a unique developmental form.

Developmental state changes how climate is expressed through growth

The same daily climate exposure can occur before a fire begins rapid expansion, during the largest growth episode, or after growth has already tapered. We therefore modeled next-day growth as a function of climate, current developmental state, and their interaction. To avoid leakage, state was defined only from information available at day t : elapsed day, current daily growth, current cumulative area, and current acceleration. Final duration, final area fraction, and future VASE coordinates were not used.

State-containing models outperform climate-only baselines for next-day growth. The best conservative state model is **core climate-state interaction**, with median held-out R^2 of 0.353. Core climate-state interactions survive the predeclared blocked-transfer margin. The interpretation is therefore cautious: developmental state improves climate interpretation and near-term growth prediction, but the present centroid climate product is not enough to claim causal state-dependent climate control.

Climate organizes opportunity without uniquely determining outcome

The closing analysis asks where climate explanation fails. Pairs of fires with similar centroid climate summaries can have divergent VASE morphologies, and pairs with similar VASE morphologies can occur under contrasting climate pathways. These mismatches are not artifacts to hide; they define the scientific boundary of the current analysis. Climate describes developmental opportunity, while active-edge exposure, local fuels, topography, vegetation, suppression, ignition context, and wind direction likely help determine which opportunity is realized.

Discussion

The main result is not that Fire VASE provides a low-dimensional coordinate system. The main result is that Fire VASE makes it possible to see climate as a probabilistic organizer of wildfire development. Climate shifts developmental-neighborhood prevalence and profile allocation, but it does not uniquely determine form.

This framing changes how climate-fire relationships should be read. Event means are informative but blunt. Daily exposure, extreme-day fractions, and developmental timing sharpen interpretation, yet transfer across regions and years remains weak. Expanded centroid climate adds moisture, fuel, and fire-danger context, but it does not remove the need for spatially

resolved exposure. Scaling perimeter and active-edge attribution, adding true local climate normals, and including topography, vegetation, suppression, ignition context, wind direction, and gusts are the next necessary steps.

The present analyses are associational. They do not isolate causal climate effects, suppression decisions, or fuel continuity. They also use daily centroid climate rather than active-edge weather, so they can miss within-perimeter heterogeneity and directional wind effects. Even with those caveats, the central claim is supported: wildfire development occupies recurring forms whose probabilities shift with climate, while realized form remains contingent on state and context.

Materials and Methods

Fire histories were read from the repository's FIRED-derived daily VASE slice table, covering 278,569 events and 626,102 daily slices from 2000-11-02 to 2021-05-01. Climate exposure was read from the full-population climate-enhanced slice table. Complete daily centroid climate values were available for 237,235 fires. Variables were maximum temperature in degrees C, minimum temperature in degrees C, VPD in kPa, wind speed in m s⁻¹, precipitation in mm d⁻¹, maximum and minimum relative humidity in percent, specific humidity in kg kg⁻¹, 100-hour and 1000-hour fuel moisture in percent, energy release component, burning index, reference evapotranspiration in mm d⁻¹, potential evapotranspiration in mm d⁻¹, and solar radiation in W m⁻². Event-level climate summaries included means, daily minima and maxima, extreme-day fractions, early/middle/late developmental-time means, and a region-month fire-season anomaly diagnostic.

Developmental response variables were defined before model fitting and separated into absolute-scale outcomes, shape-normalized responses, and time-varying state variables. Event-level models used ridge-regularized linear baselines with fixed random seed 20260722. Validation used random fire splits as a diagnostic and year, region, and region-year blocking as conservative transfer tests. State-dependent models predicted next-day growth, $\log(1 + \text{km}^2)$, using climate at day t and leakage-safe state variables available by day t . All analyses are exploratory baselines rather than causal estimates.

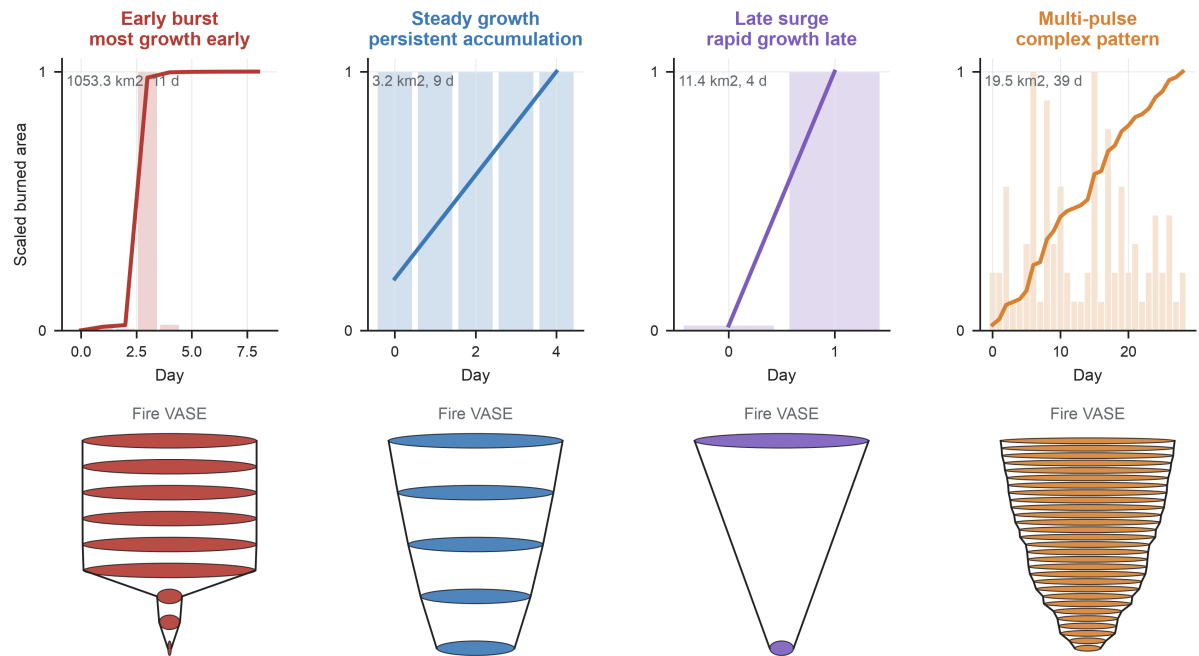
References

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Figure 1

1 Every fire has a life history.

Fires with simple final summaries can develop through different sequences.



Daily increments become ordered rings; ring width tracks cumulative burned area through developmental time.

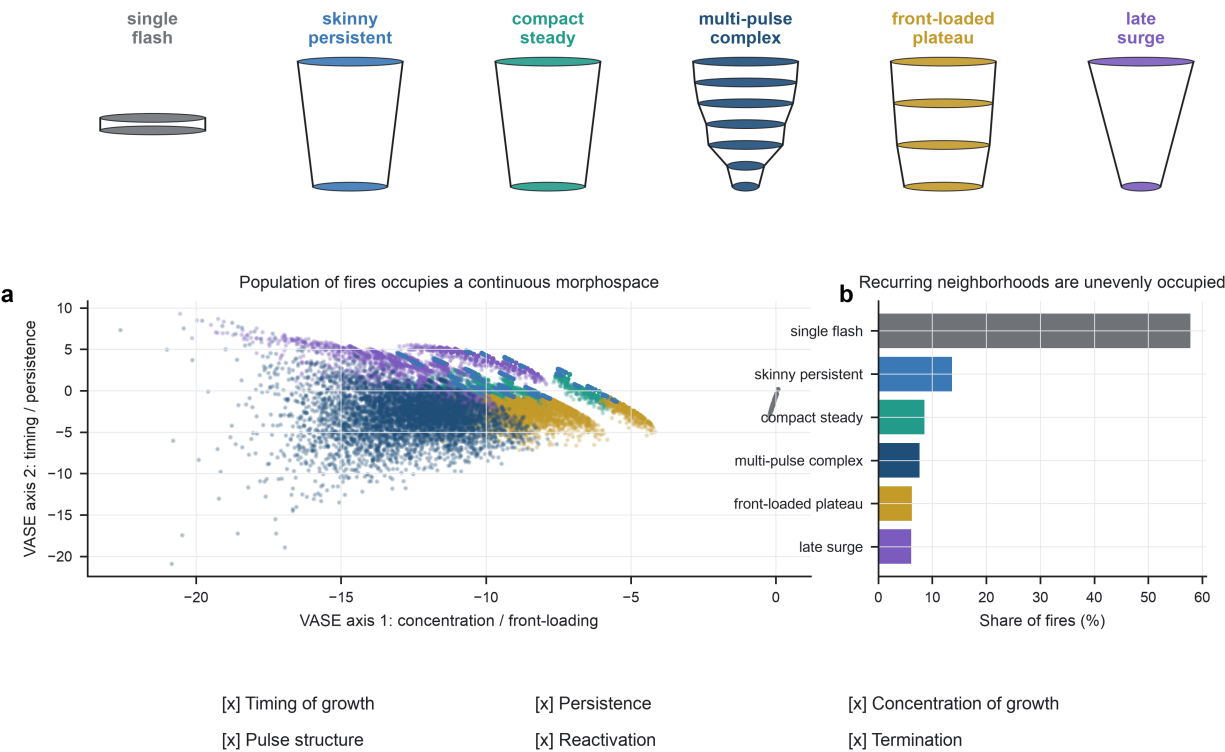
Takeaway: final area and duration hide when growth occurs; Fire VASE preserves the developmental sequence.

Four real FIRED events illustrate distinct life-history patterns: early burst, steady growth, late surge, and multi-pulse development. Bars show daily burned area scaled within fire, lines show cumulative area scaled within fire, and VASE ring width shows normalized cumulative burned area through developmental time. The figure establishes the opening premise: final area and duration are necessary descriptors but do not preserve when growth occurs.

Figure 2

2 Wildfire histories occupy recurring developmental gradients.

Fires follow continuous patterns in timing, persistence, pulse structure, reactivation, and termination.



Takeaway: development varies along continuous gradients, not as hard discrete types.

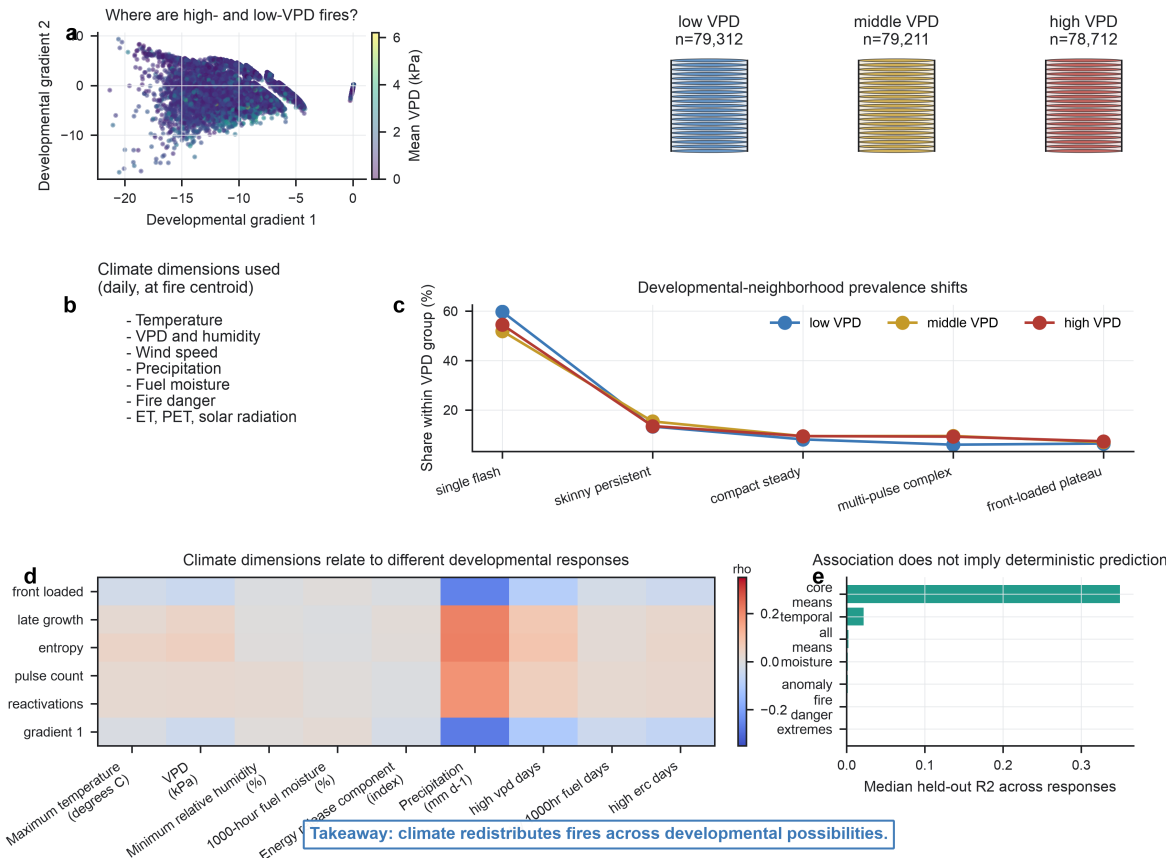
Observed representative fires mark recurring developmental neighborhoods. VASEs are labeled with interpretable shape descriptors rather than medoid IDs. The morphospace panel shows that labels grade continuously rather than forming isolated classes, the prevalence panel gives the population share of each neighborhood, and the bottom annotation names the developmental gradients that the axes summarize. The figure supports a descriptive vocabulary of timing, persistence, concentration, pulse structure, reactivation, and termination.

Figure 3

3

Climate shifts the probability of developmental forms.

Many climate dimensions are associated with where fires fall in VASE space, but they do not determine a single pathway.

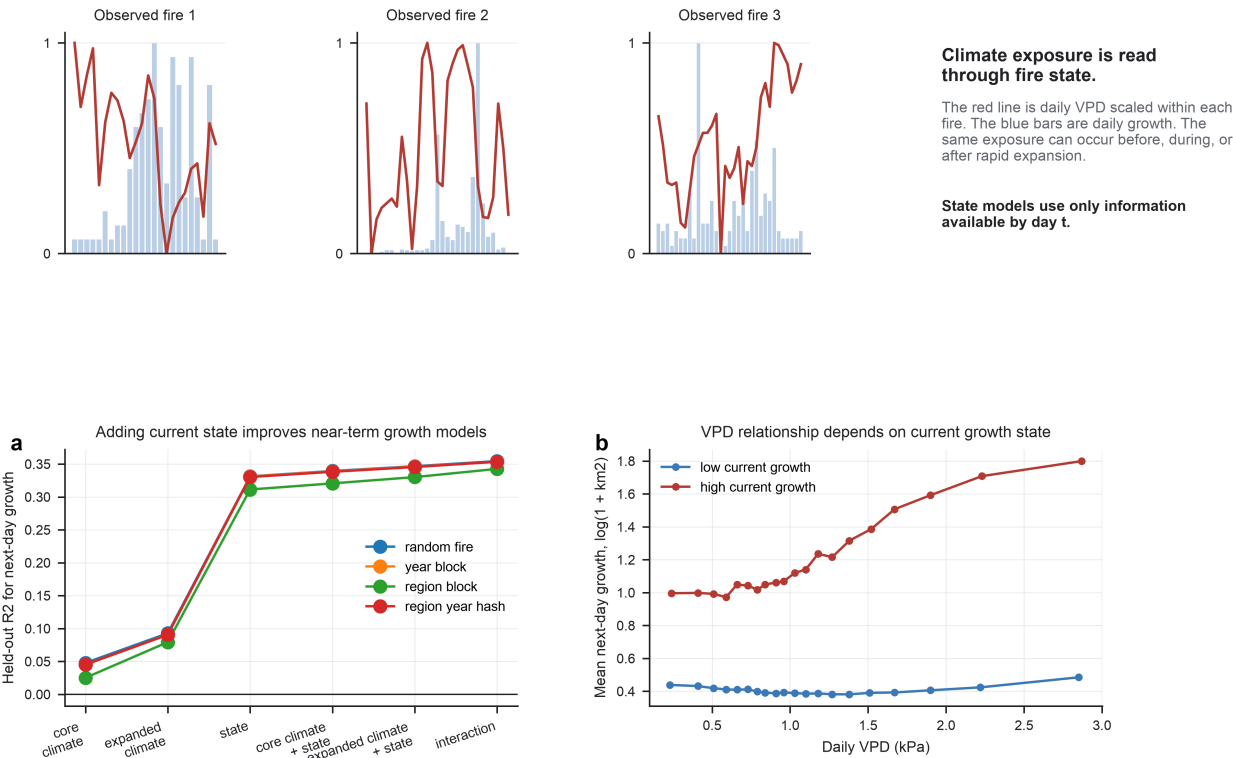


Daily centroid gridMET climate is projected onto the developmental representation. The morphospace panel maps mean VPD in kPa across VASE axes. The composite VASEs summarize low, middle, and high event-mean VPD terciles; each composite is the median normalized profile with an interquartile shell and reports group sample size. The climate-dimension panel lists the daily centroid variables used in the analysis. The prevalence panel shows how developmental-neighborhood frequency changes across VPD groups. The effect-size heatmap reports Spearman associations for temperature, VPD, precipitation, relative humidity, fuel moisture, and fire-danger summaries against interpretable responses. The blocked validation panel compares core event means, comprehensive event means, moisture/humidity, fire-danger/energy, region-season anomaly diagnostics, extreme-day fractions, and temporally resolved exposure summaries. The strongest supported conclusion is redistribution of developmental probabilities, not deterministic prediction.

Figure 4

4 Climate acts through developmental state.

The same weather exposure can mean different things depending on when it occurs in a fire history.



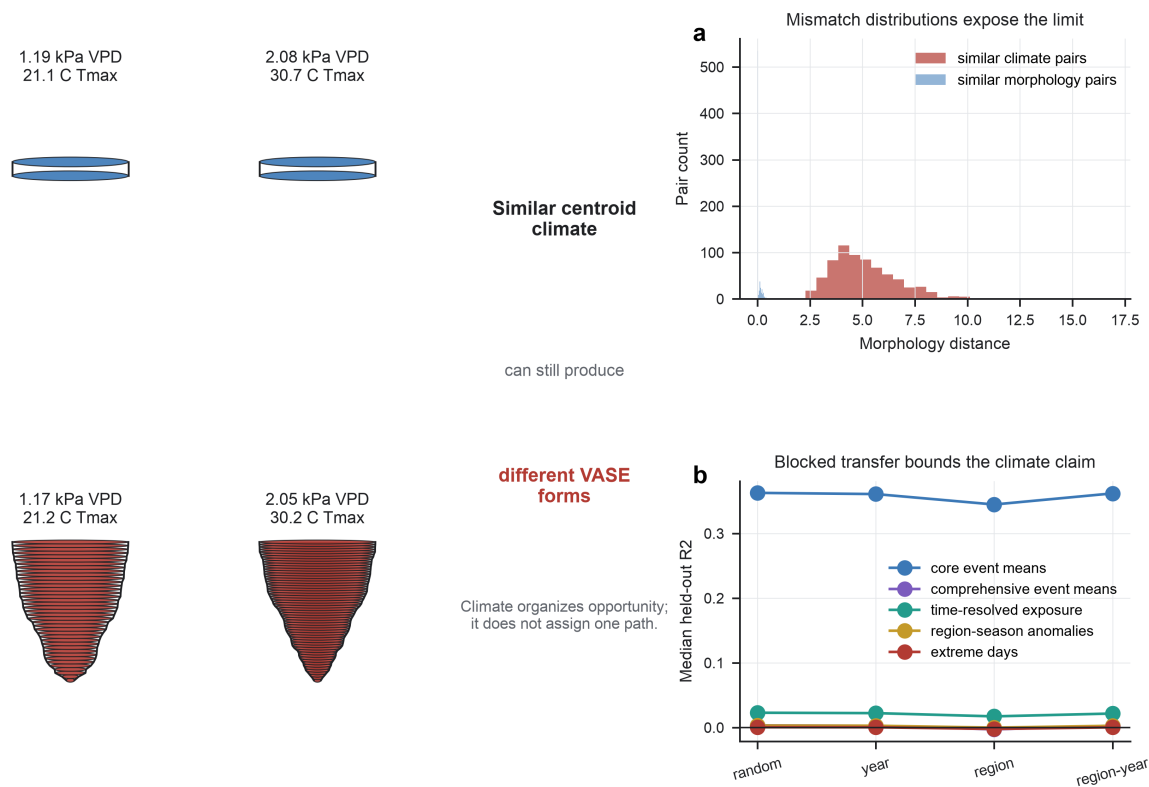
Takeaway: climate effects depend on when exposure occurs and on the current developmental state.

Representative histories align daily growth and daily VPD to show that similar exposure can occur at different developmental states. The model comparison predicts next-day growth as $\log(1 + \text{daily burned area in km}^2)$ using core climate, expanded climate, current developmental state, climate plus state, and core climate-state interactions. State predictors use only information available by day t: elapsed day, current growth, current cumulative area, and current acceleration. Conditional curves show that the VPD-growth association differs between low and high current-growth states. These are associational baselines, not causal estimates.

Figure 5

5 Climate is not destiny.

Similar climates can lead to different outcomes, and similar forms can arise under different climate pathways.



Takeaway: climate organizes opportunity, but state and landscape context help determine the path a fire takes.

The closing figure shows the limit of climate explanation. The VASE examples are pairs selected for similar centroid climate summaries but divergent developmental morphology. The central annotation states the inferential boundary: similar centroid climate can still lead to different developmental paths. Population mismatch distributions show that similar climate does not guarantee similar form, and blocked model performance shows that climate representations lose transferability across years and regions. The figure motivates missing controls that are still absent from the population table: active-edge exposure, topography, vegetation, suppression, ignition context, wind direction or gusts, and true local climate anomalies.